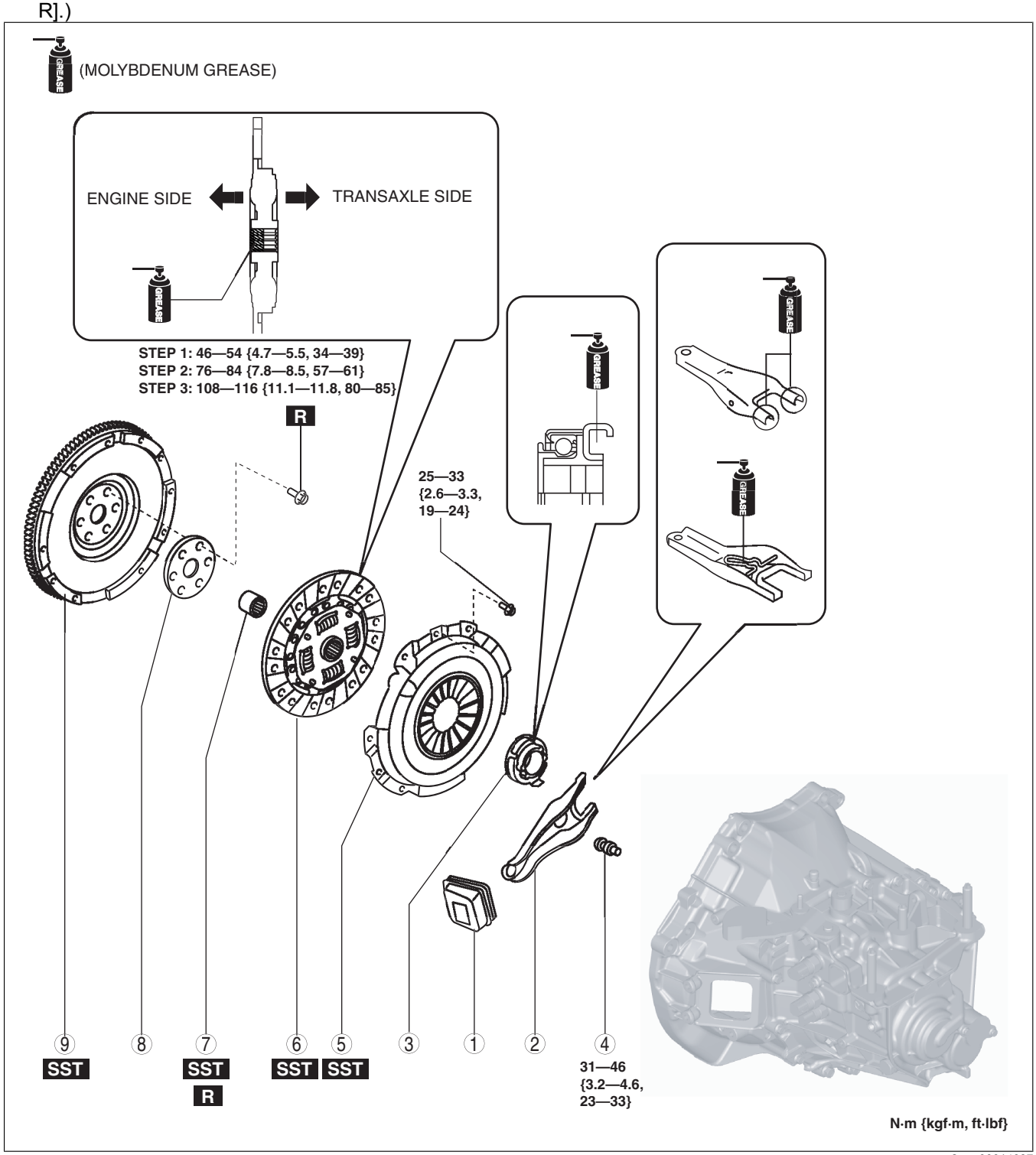
CLUTCH UNIT REMOVAL/INSTALLATION [F66M-R]

id0510q2157200

SKYACTIV-G 1.5

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the manual transaxle. (See MANUAL TRANSAXLE REMOVAL/INSTALLATION [F66M-R (SKYACTIV-G 1.5)].)
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.
5. Add the specified amount of manual transaxle oil. (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].)



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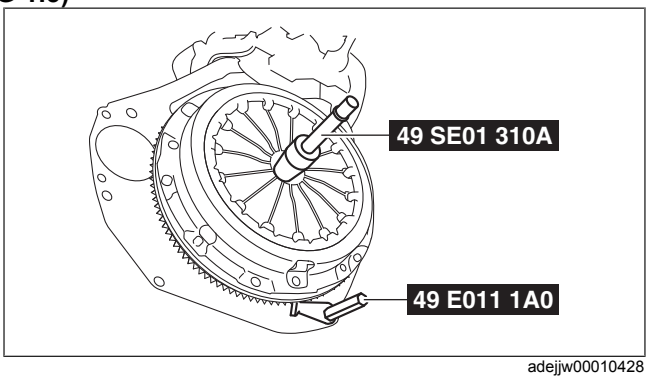
1	Boot
2	Clutch release fork
3	Clutch release collar
4	Pivot pin

5	Clutch cover (See Clutch cover and clutch disc removal note (SKYACTIV-G 1.5).) (See Clutch disc and clutch cover installation note (SKYACTIV-G 1.5).)
6	Clutch disc (See Clutch cover and clutch disc removal note (SKYACTIV-G 1.5).) (See Clutch disc and clutch cover installation note (SKYACTIV-G 1.5).)

7	Pilot bearing (See Pilot bearing removal note (SKYACTIV-G 1.5).) (See Pilot bearing installation note (SKYACTIV-G 1.5).)
8	Plate
9	Flywheel (See Flywheel removal note (SKYACTIV-G 1.5).) (See Flywheel installation note (SKYACTIV-G 1.5).)

Clutch cover and clutch disc removal note (SKYACTIV-G 1.5)

1. Hold the clutch unit using the **SSTs (49 SE01 310A, 49 E011 1A0)**.
2. Loosen each bolt one turn at a time in a crisscross pattern until spring tension is released.
3. Remove the clutch cover and clutch disc.

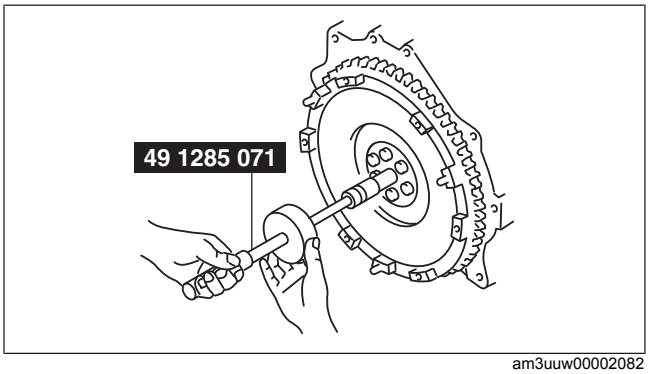


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Pilot bearing removal note (SKYACTIV-G 1.5)

Note
- The pilot bearing does not need to be removed unless you are replacing it.

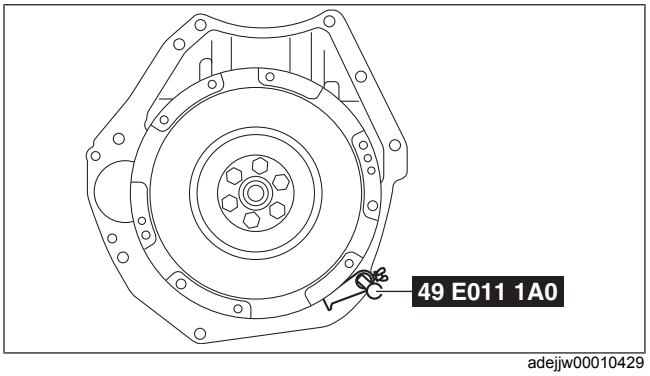
1. Use the **SST (49 1285 071)** to remove the pilot bearing.



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Flywheel removal note (SKYACTIV-G 1.5)

1. Hold the flywheel using the **SST (49 E011 1A0)**.
2. Remove the lock bolts, and remove the plate and the flywheel.
3. Inspect for oil leakage from the crankshaft rear oil seal.
   - If there is any malfunction, replace the crankshaft rear oil seal. (See REAR OIL SEAL REPLACEMENT [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)



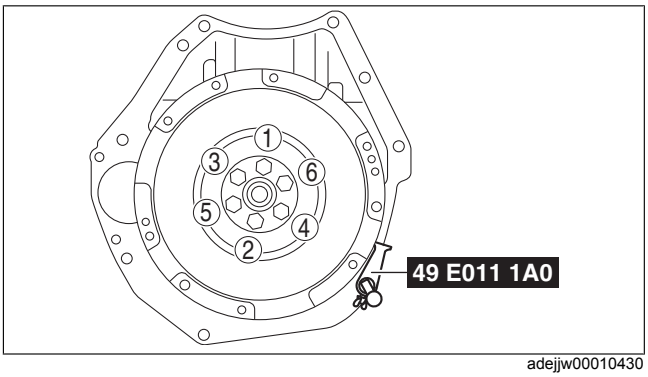
adejjw00010429

Flywheel installation note (SKYACTIV-G 1.5)

1. Degrease and clean the flywheel.
2. Clean the crankshaft thread holes.
3. Install the flywheel and the plate to the crankshaft, and temporarily tighten a new lock bolts.

4. Hold the flywheel using the **SST (49 E011 1A0)**.
5. Tighten the lock bolts completely in the order shown in the figure in the following 3 steps.

Tightening torque
Step 1: 46—54 N·m {4.7—5.5 kgf·m, 34—39 ft·lbf}
Step 2: 76—84 N·m {7.8—8.5 kgf·m, 57—61 ft·lbf}
Step 3: 108—116 N·m {11.1—11.8 kgf·m, 80—85 ft·lbf}



adejjw00010430

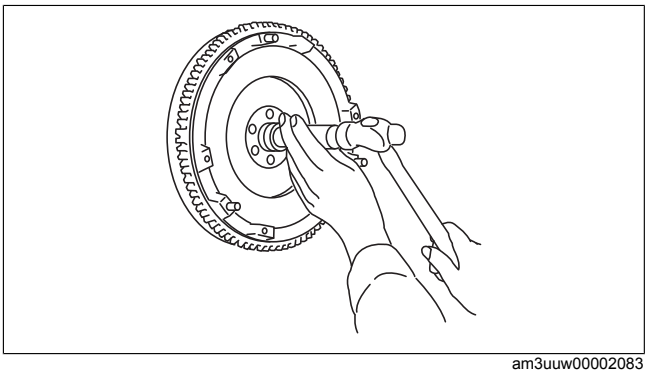
Pilot bearing installation note (SKYACTIV-G 1.5)

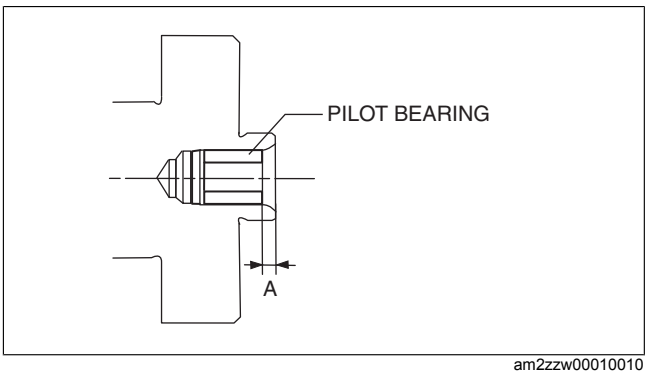
1. Install new pilot bearing to the specified position using the following tools.

Tool
Snap-on brand millimeter size bushing driver set (A160M) adapter A160M7 (20—22 mm)
- **Use the adapter with the 20 mm side of the A160M7 (20—22 mm) facing the pilot bearing side.**

Substitution tool
Outer diameter: 21 mm {0.83 in}
Inner diameter: 19 mm {0.75 in}

Specified position
Distance A of pilot bearing from crankshaft end: 1.5—2.5 mm {0.060—0.098 in}


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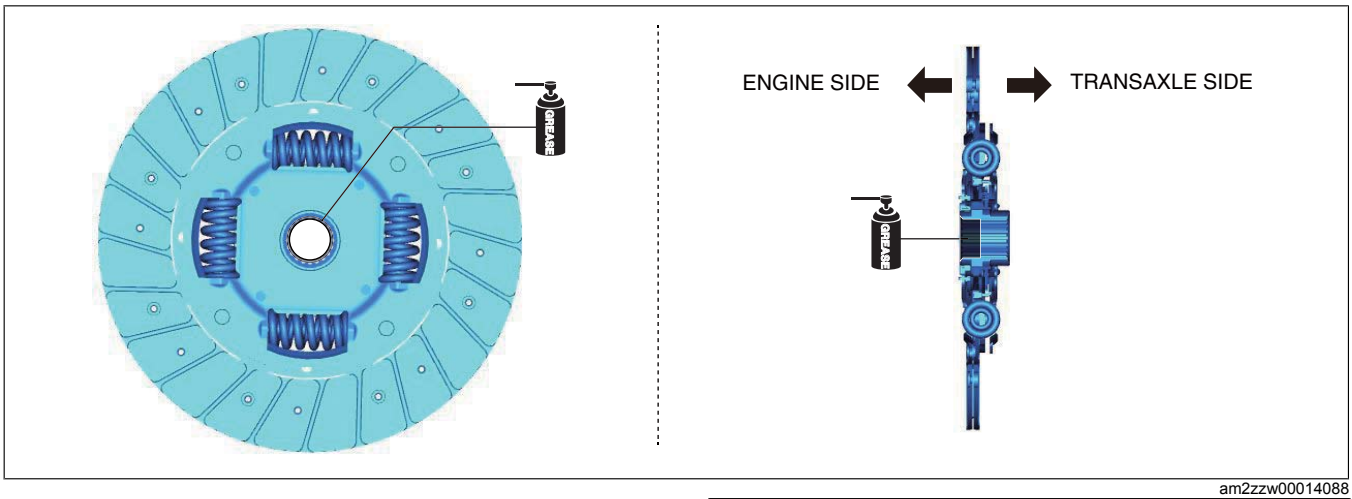


am2zzw00010010

Clutch disc and clutch cover installation note (SKYACTIV-G 1.5)

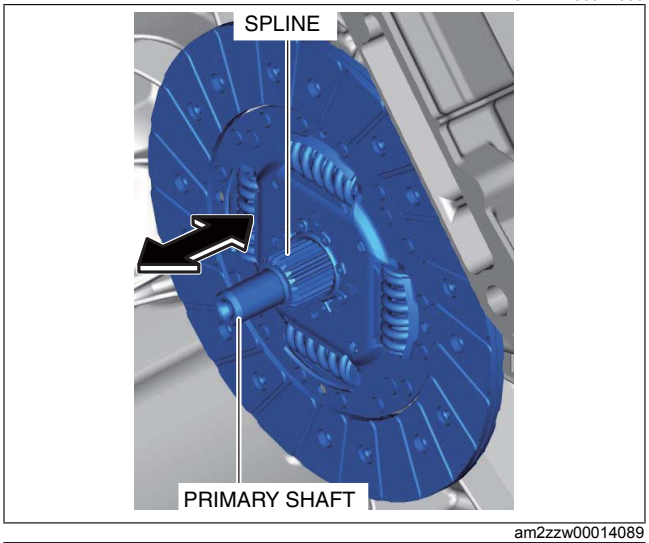
Caution
- **Do not apply grease to the spline end (transaxle side). If excessive grease is applied, clutch-slip may occur.**

1. Apply grease to the spline (engine side) of the clutch disc.



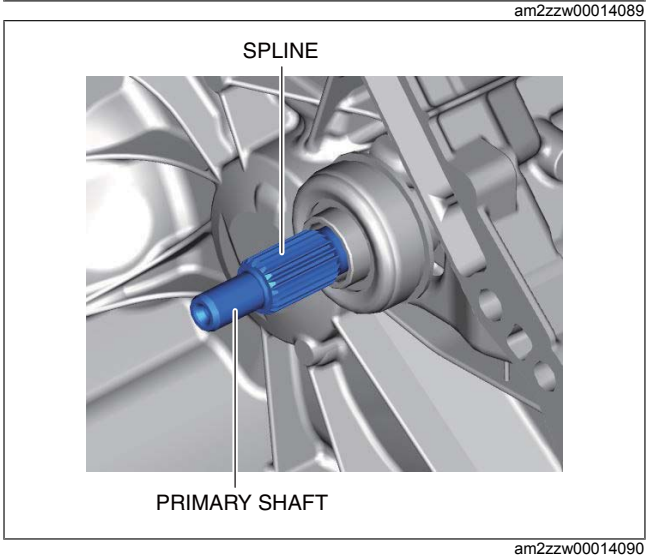
am2zzw00014088

2. Insert the clutch disc into the primary shaft and wipe off the excess grease protruding from the spline.
3. Slide the clutch disc three times in the directions of the arrows shown in the figure to engage to grease.



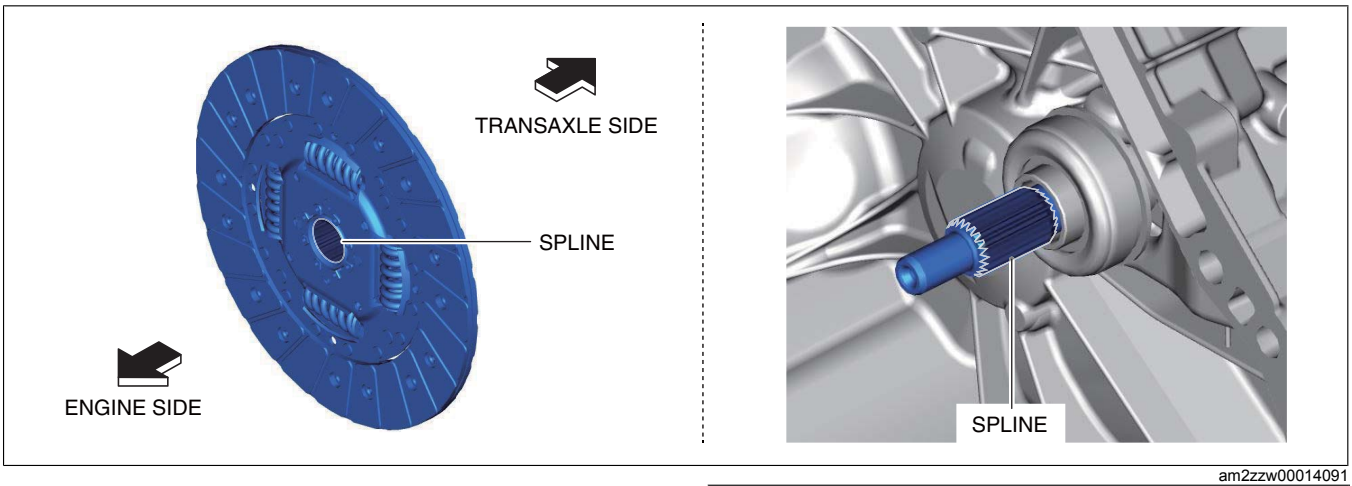
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4. Remove the clutch disc from the primary shaft and wipe off the excess grease protruding from the spline.



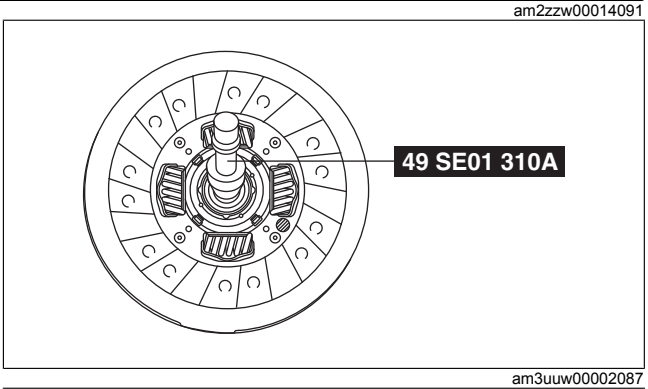
am2zzw00014090

5. Verify that grease has been lightly applied to the clutch disc and the spline of the primary shaft.



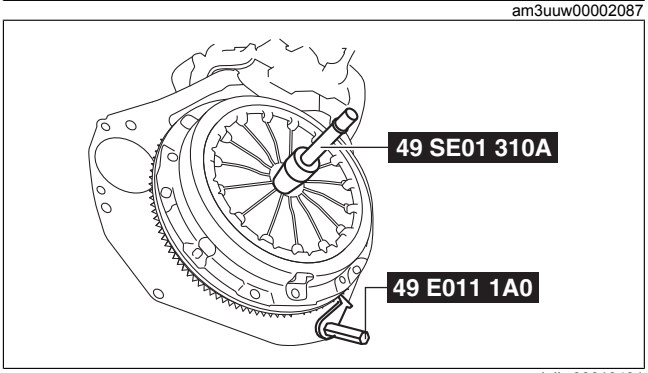
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6. Hold the clutch disc position using the **SST (49 SE01 310A)**.



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7. Hold the clutch unit using the **SST (49 E011 1A0)**.

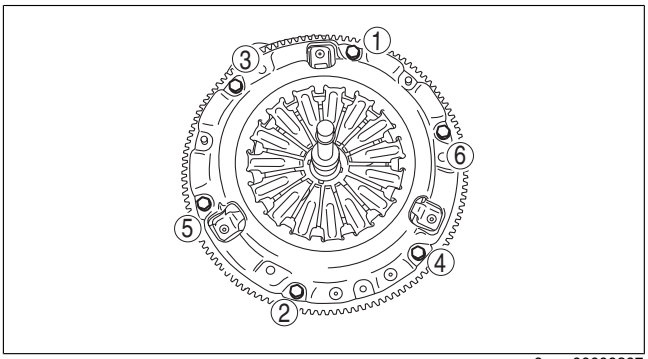


adejjw00010431

8. Using the following procedure, install the clutch disc and clutch cover.
   (1) Temporarily tighten the bolts in the order shown in the figure.
   (2) Tighten the bolts completely in two or three steps in the order shown in the figure and install the clutch disc and cover.

   Tightening torque
   25—33 N·m {2.6—3.3 kgf·m, 19—24 ft·lbf}

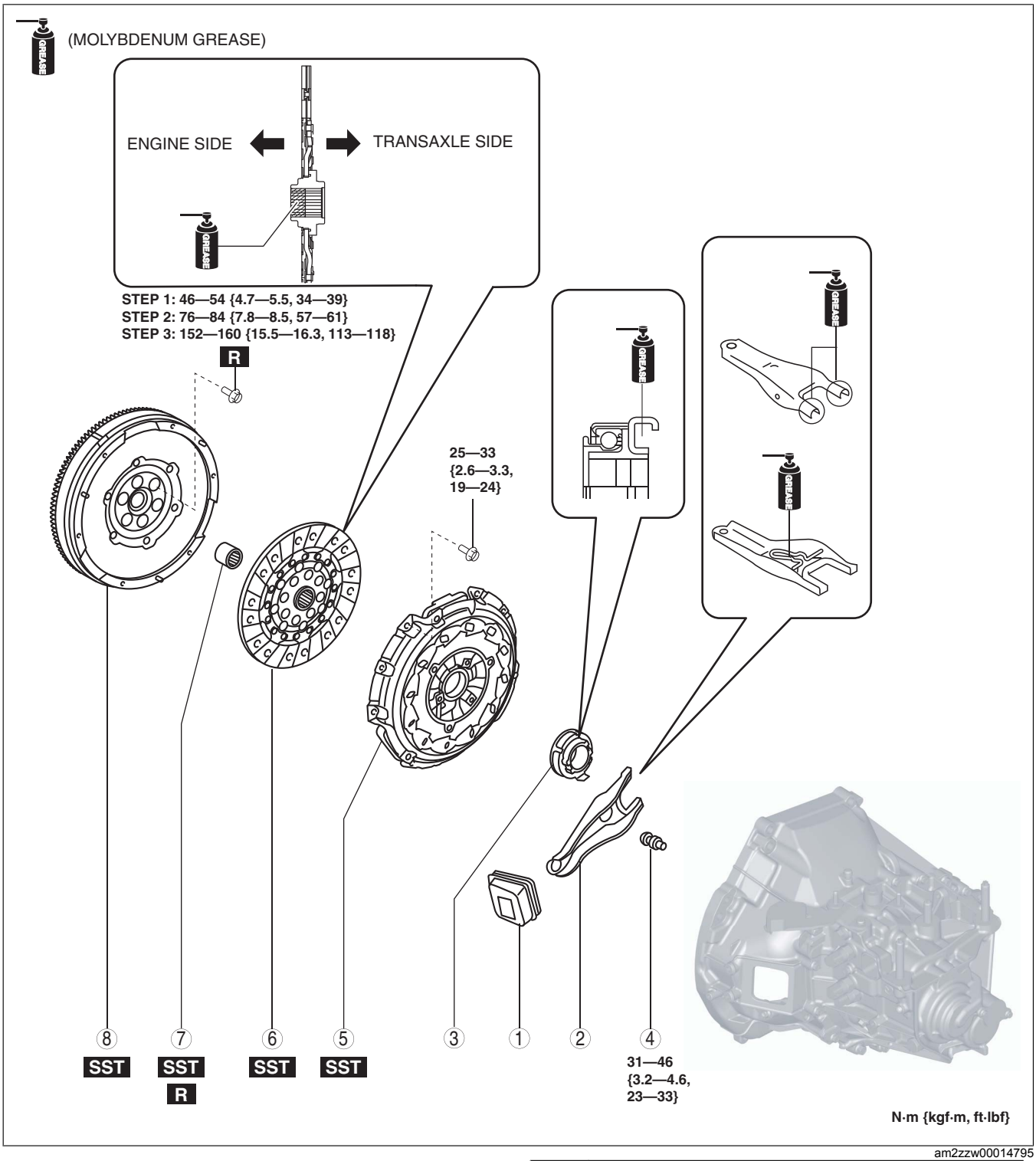
9. Remove the **SSTs (49 SE01 310A, 49 E011 1A0)**.



am3uuw00008267

SKYACTIV-D 1.5

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the manual transaxle. (See MANUAL TRANSAXLE REMOVAL/INSTALLATION [F66M-R (SKYACTIV-D 1.5)].)
3. Remove in the order indicated in the table.
4. Install in the reverse order of removal.
5. Add the specified amount of manual transaxle oil. (See MANUAL TRANSAXLE OIL REPLACEMENT [F66M-R].)



am2zzw00014795

1	Boot
2	Clutch release fork
3	Clutch release collar
4	Pivot pin
5	Clutch cover (See Clutch cover and clutch disc removal note (SKYACTIV-D 1.5).) (See Clutch disc and clutch cover installation note (SKYACTIV-D 1.5).)

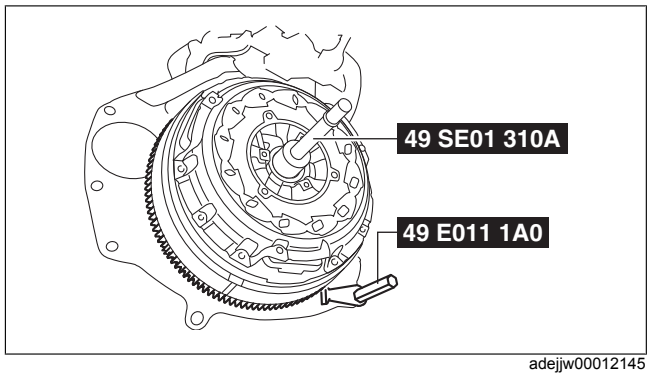
6	Clutch disc (See Clutch cover and clutch disc removal note (SKYACTIV-D 1.5).) (See Clutch disc and clutch cover installation note (SKYACTIV-D 1.5).)
7	Pilot bearing (See Pilot bearing removal note (SKYACTIV-D 1.5).) (See Pilot bearing installation note (SKYACTIV-D 1.5).)

8	Dual-mass flywheel (See Dual-mass flywheel removal note (SKYACTIV-D 1.5).) (See Dual-mass flywheel installation note (SKYACTIV-D 1.5).)

Clutch cover and clutch disc removal note (SKYACTIV-D 1.5)

Caution
- **A function for eliminating set load changes, which occur due to clutch disc wear, is equipped in the clutch cover of the F66M-R manual transaxle for the SKYACTIV-D 1.5. Because the clutch cover deforms once it is used, it can only provide a correct set load when it used with the clutch disc that is equipped with it. Therefore, if the clutch cover or clutch disc is newly replaced, perform the following.**
  - **— When newly replacing the clutch cover, also newly replace the clutch disc.**
  - **— When newly replacing the clutch disc, also newly replace the clutch cover.**

1. Hold the clutch unit using the **SSTs (49 SE01 310A, 49 E011 1A0)**.
2. Loosen each bolt one turn at a time in a crisscross pattern until spring tension is released.
3. Remove the clutch cover and clutch disc.

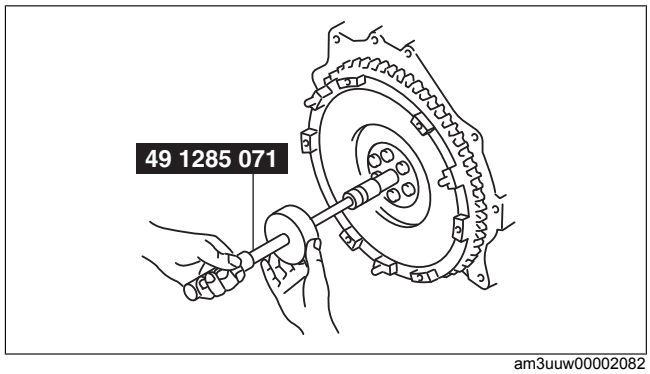


adejjw00012145

Pilot bearing removal note (SKYACTIV-D 1.5)

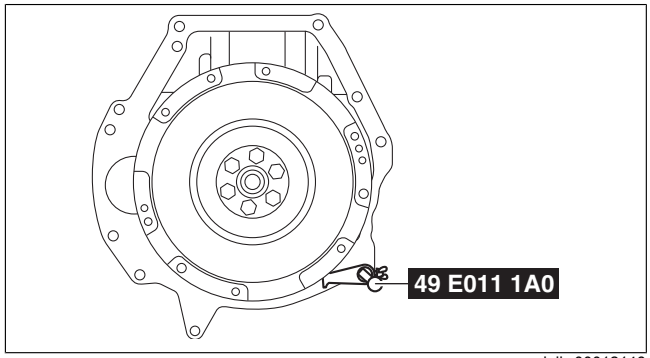
Note
- The pilot bearing does not need to be removed unless you are replacing it.

1. Use the **SST (49 1285 071)** to remove the pilot bearing.



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Dual-mass flywheel removal note (SKYACTIV-D 1.5)
1. Hold the dual-mass flywheel using the **SST (49 E011 1A0)**.
2. Remove the lock bolts, and remove the plate and the dual-mass flywheel.
3. Inspect for oil leakage from the crankshaft rear oil seal.
   - If there is any malfunction, replace the crankshaft rear oil seal. (See REAR OIL SEAL REPLACEMENT [SKYACTIV-D 1.5].)



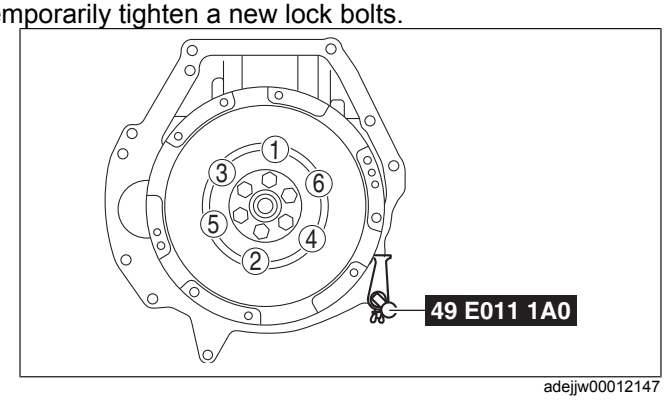
adejjw00012146

Dual-mass flywheel installation note (SKYACTIV-D 1.5)

Caution
- **When cleaning the dual-mass flywheel, use a dry cloth with no fluid.**
- **Do not clean between the primary flywheel and secondary flywheel. Clean only the bolt connecting areas and the contact surface of the clutch.**

1. Clean the dual-mass flywheel.
2. Clean the crankshaft thread holes.
3. Install the dual-mass flywheel to the crankshaft, and temporarily tighten a new lock bolts.
4. Hold the dual-mass flywheel using the **SST (49 E011 1A0)**.
5. Tighten the lock bolts completely in the order shown in the figure in the following 3 steps.

Tightening torque
Step 1: 46—54 N·m {4.7—5.5 kgf·m, 34—39 ft·lbf}
Step 2: 76—84 N·m {7.8—8.5 kgf·m, 57—61 ft·lbf}
Step 3: 152—160 N·m {15.5—16.3 kgf·m, 113—118 ft·lbf}



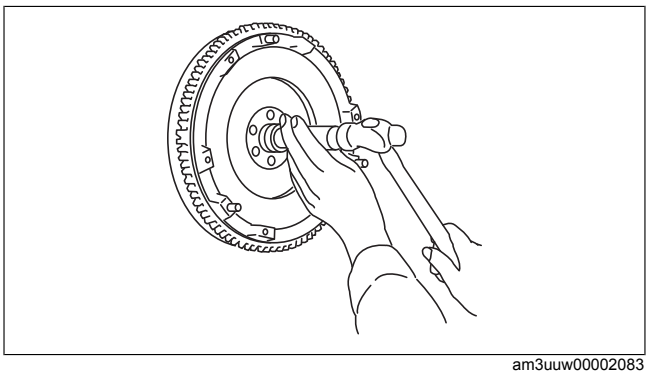
adejjw00012147

Pilot bearing installation note (SKYACTIV-D 1.5)

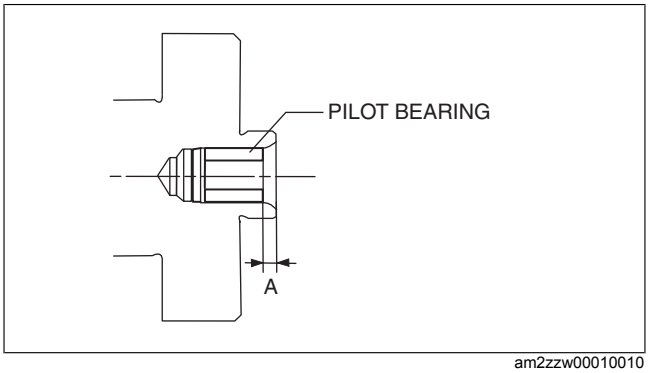
1. Install new pilot bearing to the specified position using the following tools.

Tool
Snap-on brand millimeter size bushing driver set (A160M) adapter A160M7 (20—22 mm)
- **Use the adapter with the 20 mm side of the A160M7 (20—22 mm) facing the pilot bearing side.**

Substitution tool
Outer diameter: 21 mm {0.83 in}
Inner diameter: 19 mm {0.75 in}


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Specified position
Distance A of pilot bearing from crankshaft end: 1.5—2.5 mm {0.060—0.098 in}

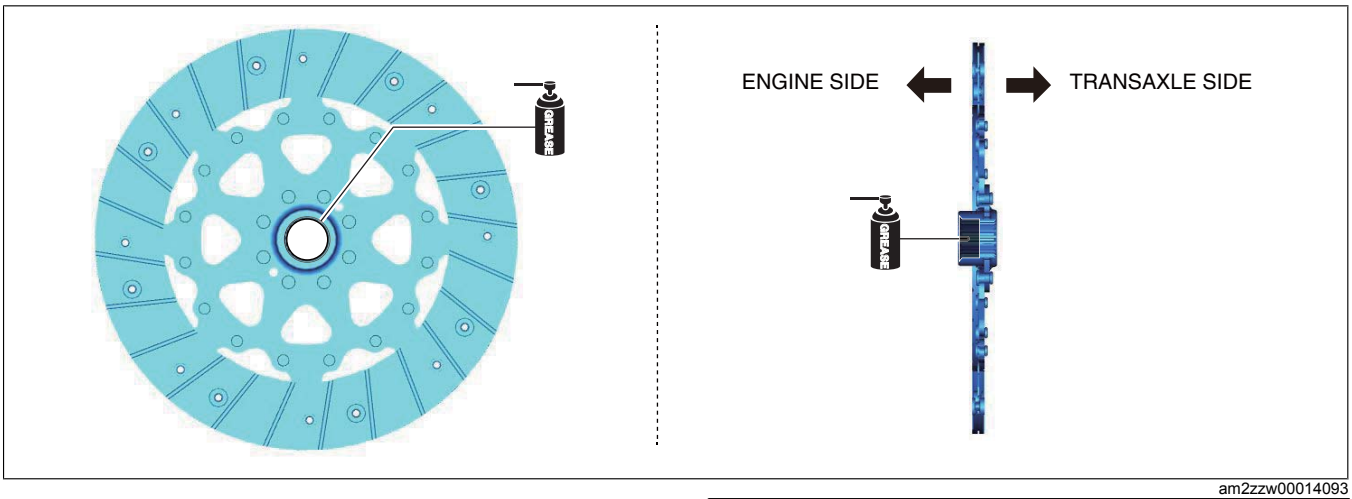


am2zzw00010010

Clutch disc and clutch cover installation note (SKYACTIV-D 1.5)

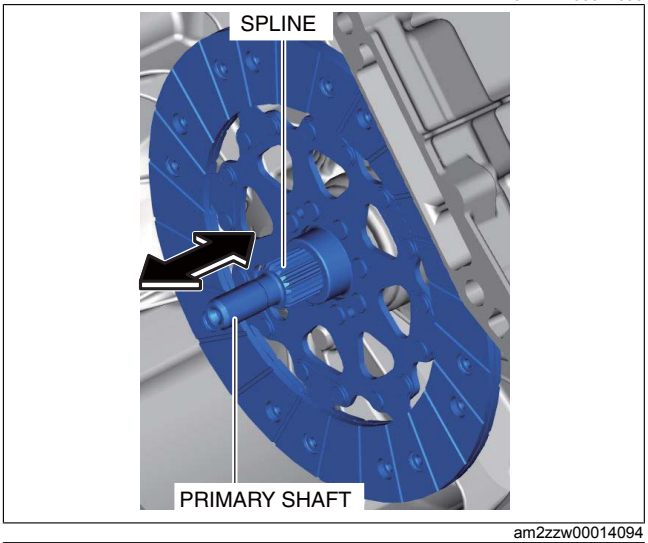
Caution
- **Do not apply grease to the spline end (transaxle side). If excessive grease is applied, clutch-slip may occur.**

1. Apply grease to the spline (engine side) of the clutch disc.



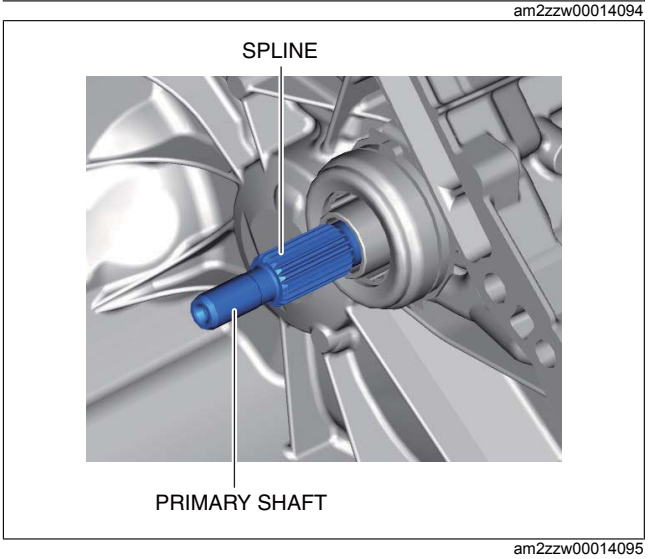
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2. Insert the clutch disc into the primary shaft and wipe off the excess grease protruding from the spline.
3. Slide the clutch disc three times in the directions of the arrows shown in the figure to engage to grease.



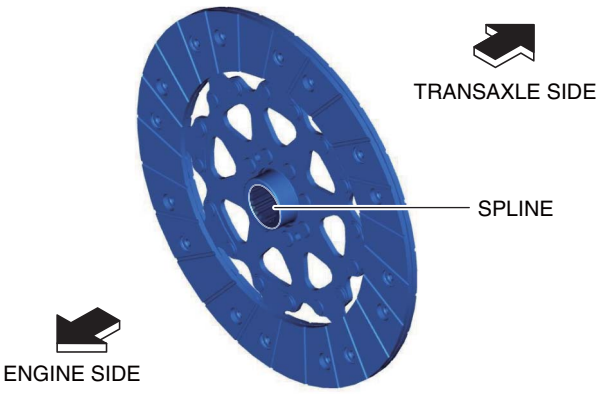
am2zzw00014094

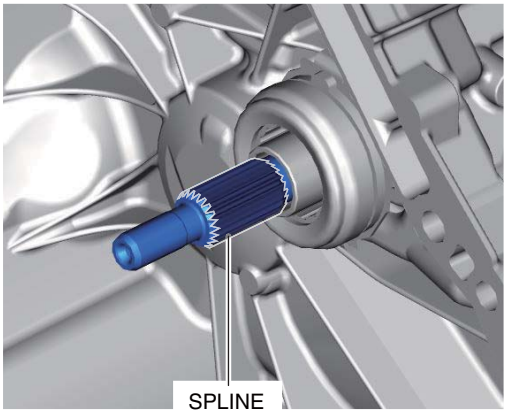
4. Remove the clutch disc from the primary shaft and wipe off the excess grease protruding from the spline.



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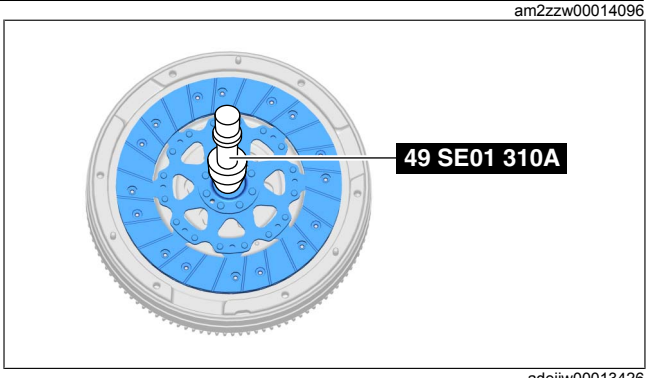
5. Verify that grease has been lightly applied to the clutch disc and the spline of the primary shaft.





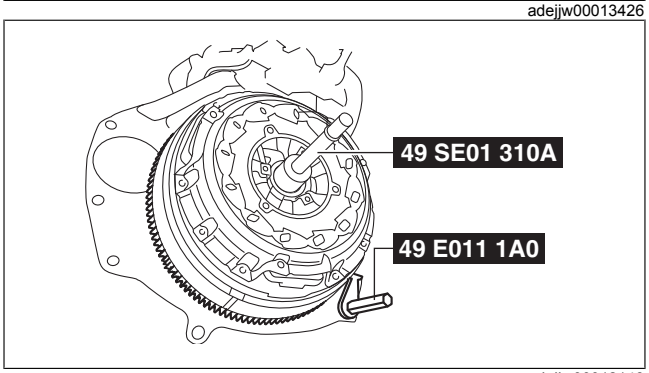
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6. Hold the clutch disc position using the **SST (49 SE01 310A)**.



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7. Hold the clutch unit using the **SST (49 E011 1A0)**.

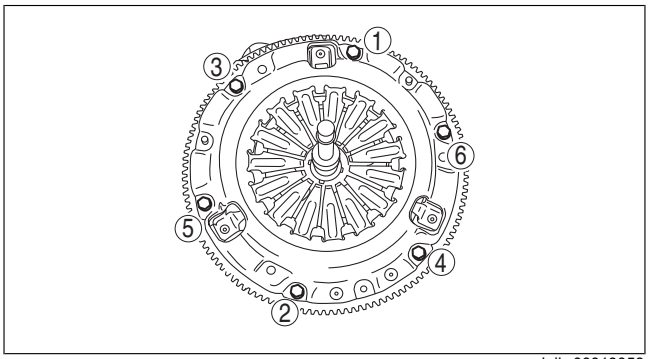


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8. Using the following procedure, install the clutch disc and clutch cover.
   (1) Temporarily tighten the bolts in the order shown in the figure.
   (2) Tighten the bolts completely in two or three steps in the order shown in the figure and install the clutch disc and cover.

   Tightening torque
   25—33 N·m {2.6—3.3 kgf·m, 19—24 ft·lbf}

9. Remove the **SSTs (49 SE01 310A, 49 E011 1A0)**.



adejjw00013058